FLAGSHIP 02 · BIONIC EXOSKELETON SERIES

LC Flex-W

Flexible Waist Exoskeleton · Unpowered Smart Lumbar Belt

The Unpowered Smart Belt. Up to 30kg Dynamic Lumbar Thrust.

This lightweight, unpowered lumbar exoskeleton is designed for frequent bending and lifting tasks. It stores and releases energy via elastic material deformation, acting as a biomechanical "smart belt" to deliver upward assistive thrust during bending. Significantly reduces lumbar strain, offloads spinal disc compression, and mitigates occupational lower-back injury risks.



01 Sub-1kg Minimalist Chassis

Breathable textile construction conforms naturally to the torso, guaranteeing sweat-free comfort throughout an 8-hour shift.

02 Unrestricted Agility

Conforms to body contours without rigid metal hip constraints, preserving full freedom for walking, crouching, climbing, and driving.

03 Dynamic Erector Spinae Offloading

Progressively builds elastic tension as the user bends forward, releasing explosive upward thrust upon standing.

04 Spinal Pressure Equilibrium

Distributes acute lumbar point loads across the thoracic spine and thighs, preventing occupational disc herniation.

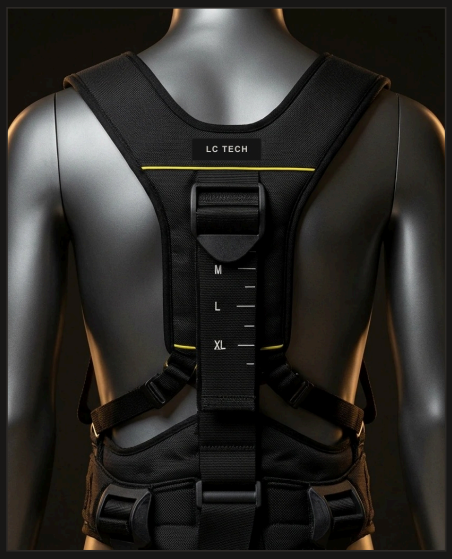
KEY METRIC 01
0-30kg
Dynamic Upward Lumbar Assist Range

KEY METRIC 02
<= 1kg
Ultra-Slim Wearable Profile

KEY METRIC 03
160-190cm
Broad Ergonomic Height Adaptation

KEY METRIC 04
0V
Non-Powered Fail-Safe Kinetic Energy Storage

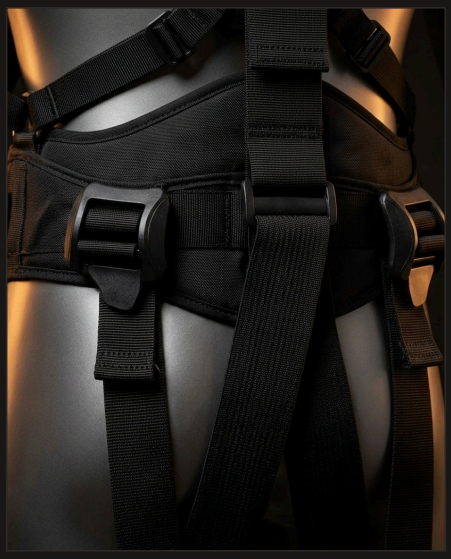
MATERIAL SCIENCE & MICRO-ARCHITECTURE · 3200K STUDIO MACRO INSPECTION



3D BREATHABLE MESH + LUMBAR BRACKET

Integrated Back-Lumbar Support Bracket & 3D Cushion

An ergonomic multi-layer 3D honeycomb lumbar cushion houses the integrated back-lumbar support bracket, stabilizing the lower spine while venting heat and moisture during high-frequency warehouse sorting.



HIGH-RESILIENCE POLYESTER LATEX

High-Resilience Polyester Latex Fiber Core

Engineered elastic polyester latex fibers store mechanical potential energy during forward trunk flexion and return up to 30kg of smooth upward assistive thrust.



WEAR-RESISTANT TACTICAL OXFORD

Bilateral Thigh Linkage Strap System

Soft wear-resistant Oxford thigh cuffs anchor the kinetic load pathway without rigid metal hip hinges, allowing operators to transition directly from lifting to driving forklifts.

ANATOMICAL & MECHANICAL SUBSYSTEMS (5 NODES)

1. **Multi-Height Adjustable Back [Thoracic Tension Anchor]**
   Telescoping shoulder tension system accommodates operator heights from 160cm to 190cm.
2. **Flank Articulation Routing [Lateral Oblique Guide]**
   Soft flexible strap pathways eliminate hip pinching during extreme deep squats.
3. **Thickened 3D Lumbar Cushion [L1-L5 Decompression Pad]**
   Ergonomic multi-layer pad cushions the lower spine and distributes back loads.
4. **Fail-Safe Elastic Core [Kinetic Deformation Band]**
   High-resilience polyester latex fibers ensure lifetime durability in wet or dusty workplaces.
5. **Bilateral Thigh Traction Straps [Femoral Linkage Cuffs]**
   Integrated lumbar support bracket linked to bilateral thigh cuffs for smooth 0-30kg load transfer.

FIELD DEPLOYMENT | SMART LOGISTICS FULFILLMENT CENTER | PALLETIZING ZONE



High-Frequency Logistics Hubs & Airport Baggage Handling

In high-throughput logistics sorting centers and airport baggage bays, handlers perform hundreds of deep bends and heavy lifts every hour. Weighing <=1kg with zero rigid hip bars, the LC Flex-W acts as a second set of erector spinae muscles - delivering up to 30kg of upward lumbar assist while permitting natural walking, climbing, and vehicle operation.

0-30kg Adjustable	<= 1.0kg	160-190cm
Dynamic Upward Assist	Total Wearable Weight	Operator Height Range

COMPLETE FACTORY TECHNICAL SPECIFICATIONS · LC Flex-W (100% UNABBREVIATED)

ENGINEERING GROUP	PARAMETER	VERIFIED FACTORY SPECIFICATION
IDENTITY & COMPLIANCE	**Model**	LC Flex-W
IDENTITY & COMPLIANCE	**Color**	Tactical Black
IDENTITY & COMPLIANCE	**Target Users**	Manual laborers, warehouse operators, logistics handlers, caregivers
IDENTITY & COMPLIANCE	**Applications**	Industrial logistics, airport baggage, medical elderly care, frequent bending operations
PERFORMANCE & MECHANICS	**Assistance Type**	Non-powered passive mechanical
DIMENSIONS & WARRANTY	**Applicable Height**	160-190cm
PERFORMANCE & MECHANICS	**Assistance Range**	0-30kg (dynamically adjustable)
MATERIALS & ERGONOMICS	**Energy Storage Medium**	High-resilience elastic polyester latex fiber
PERFORMANCE & MECHANICS	**Core Structure**	Integrated back-lumbar support bracket + bilateral thigh linkage strap system
MATERIALS & ERGONOMICS	**Outer Fabric**	Wear-resistant Oxford fabric
MATERIALS & ERGONOMICS	**Lining Fabric**	Breathable 3D mesh fabric
DIMENSIONS & WARRANTY	**Sizes**	Standard Size / Plus Size
DIMENSIONS & WARRANTY	**Net Weight**	<= 1kg
DIMENSIONS & WARRANTY	**Package Dimensions**	40 x 30 x 10cm
DIMENSIONS & WARRANTY	**Warranty**	1-Year Official Industrial Warranty

DIRECT FACTORY PROCUREMENT & PILOT DEPLOYMENT - LC Flex-W

Manufacturer: Guangdong Langcheng Technology Co., Ltd. | R&D Base: Intelligent Protection Technology R&D Center (GDUT Institute of CNC Equipment Collaborative Innovation)

Hotline / WhatsApp: +86 159-8917-5854 | Engineering Email: 1094099100@qq.com | Warranty: 1-Year Official Industrial Manufacturer Warranty Across All Hardware